

BEYOND ASSUMPTIONS: MEASURING FEDERATED LEARNING OVER REAL 5G NETWORKS

R.J. Hayek^{*12}, K. Comer^{*1}, J. Chung², C. R. Murthy³, R. Kettimuthu², I. Kadota¹
Northwestern University¹, Argonne National Laboratory², Indian Institute of Science³
^{*} These authors contributed equally to this work.

MOTIVATION

- Lack of real-world implementation (mostly theory, simulations, and emulations)
- Measurements lead to more realistic modeling
 - Validate theory and uncover unexpected relationships among variables

OUR CONTRIBUTIONS [1]

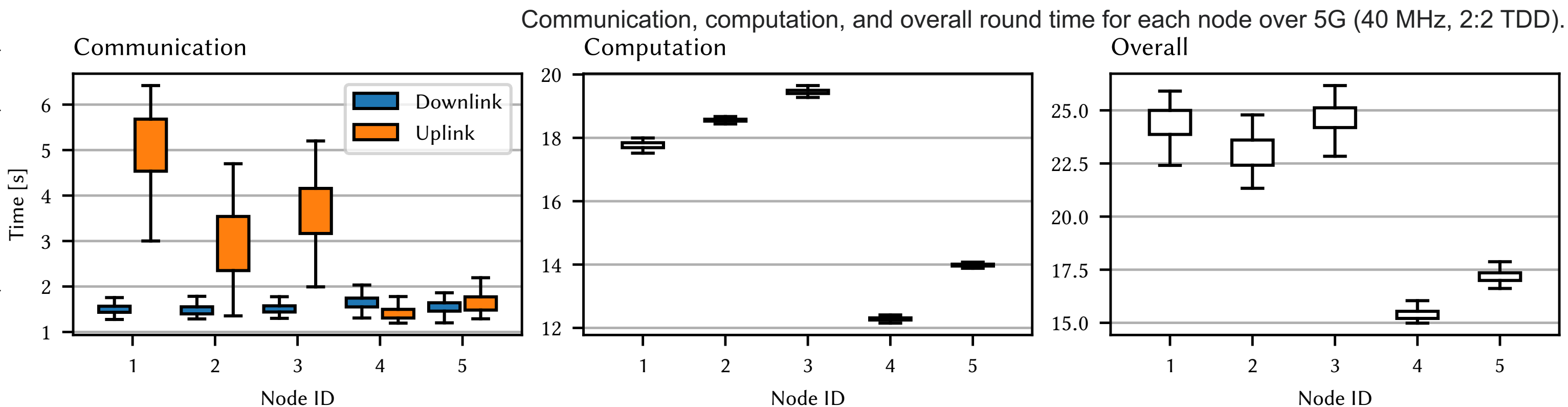
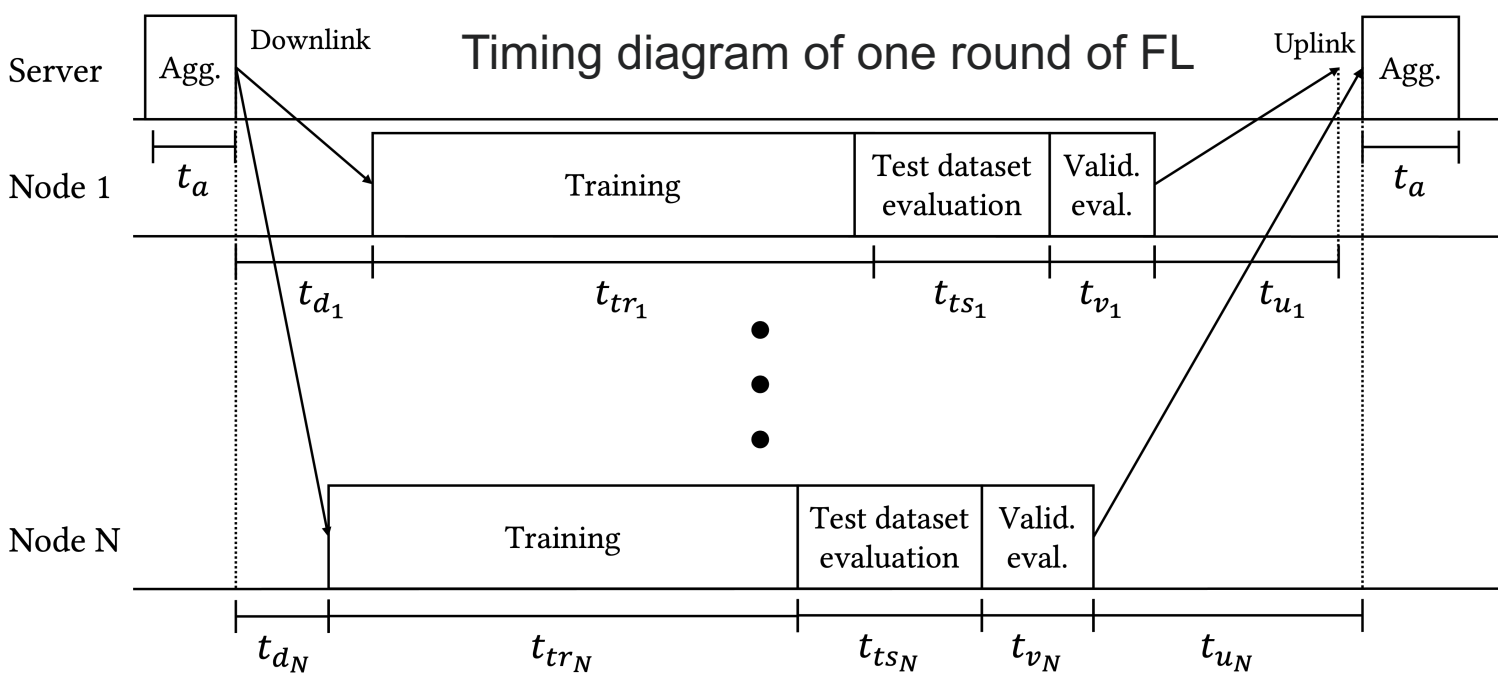
- Deploying FL on edge devices over real networks
- Compiling a comprehensive measurement dataset; releasing data and software publicly [2]
- Identifying discrepancies between common assumptions and real-world data

COMMON ASSUMPTIONS

- Common assumptions include:
 - Same communication time for each device [3,4] and over time (stationary) [5,6]
 - Computation time depends only on hardware [5,7]
- We show:
 - Communication time varies by device and over time
 - Normalized computation time varies over devices with identical hardware
 - Hardware specification does not fully predict performance

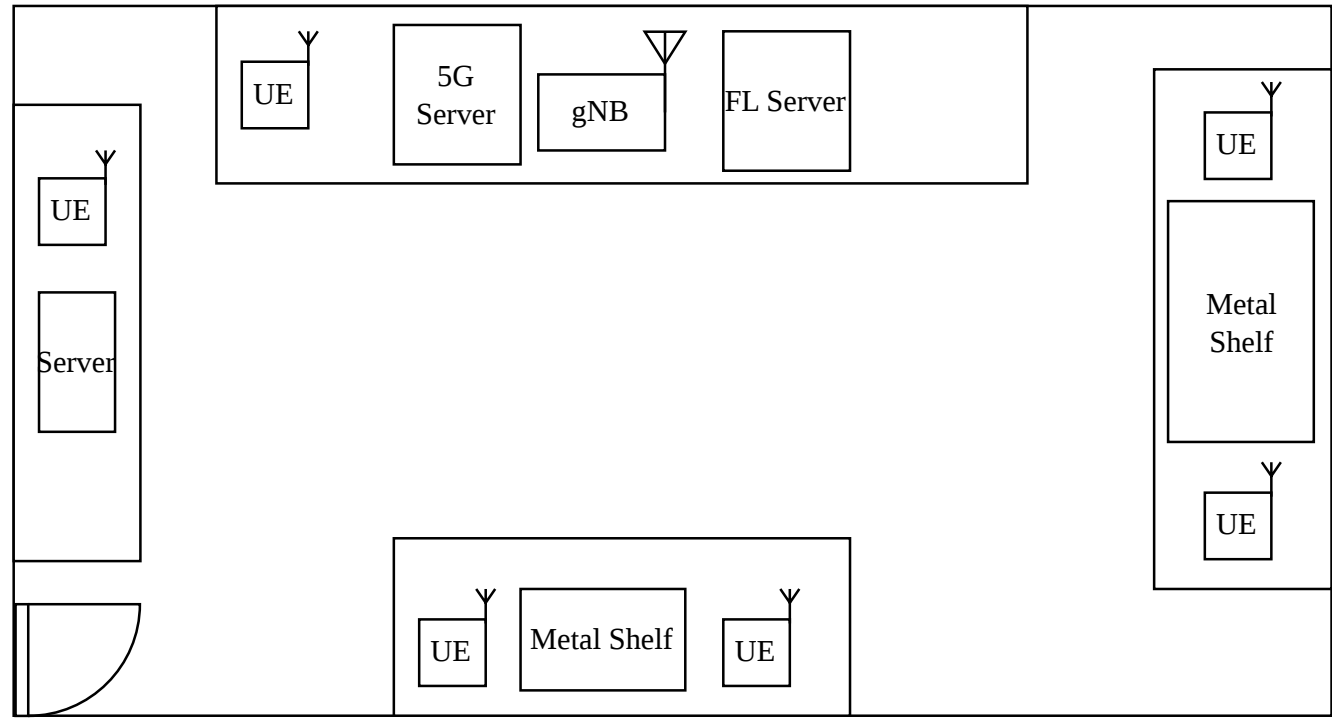
KEY TAKEAWAYS

- Real 5G violates common modeling assumptions
 - Latency is device-dependent and non-stationary
 - Training time varies among devices with identical hardware
- Stragglers are not always fixed
- Measurement-based 3GPP models outperform Shannon-style idealizations for predicting uplink time



BACKGROUND

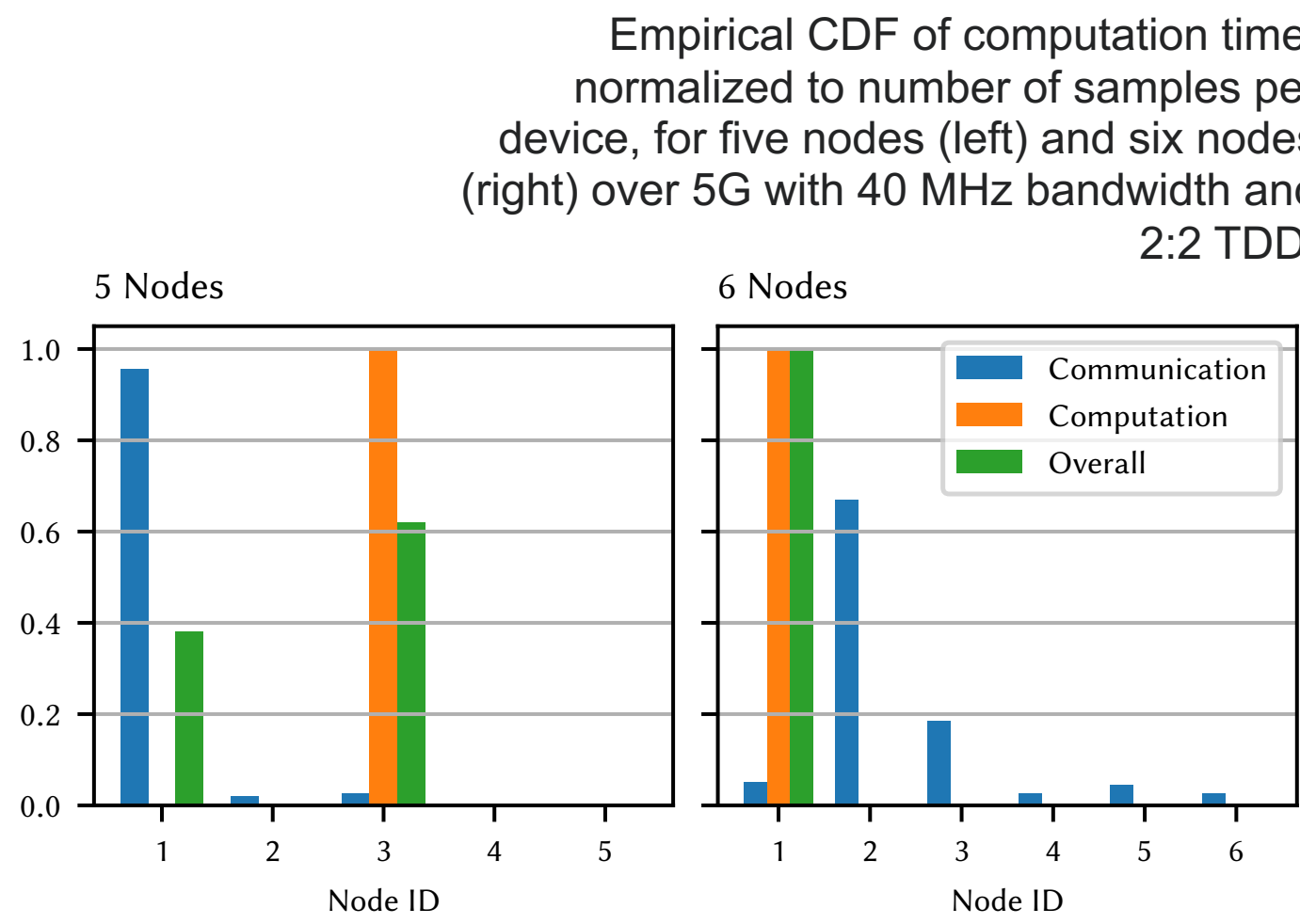
- Federated Learning (FL): distributed training of machine learning model
- 5G: latest cellular standard
 - Open-source: Open Radio Access Network
- Base station (BS) schedules communications
- Time-division duplexing (TDD) divides each transmission frame into slots for uplink (UL, UE to BS) and downlink (DL, BS to UE)



A floorplan of our FL-over-5G laboratory.

IMPLEMENTATION

- FL: Flower open-source framework
 - SqueezeNet convolutional neural net
 - Model parameter transmissions ~3 MB
 - CIFAR10 image classification dataset
 - Dirichlet-distributed
 - 200 rounds
- 150 FL-over-5G experiments; 350+ hours of measurements
 - Bandwidths: 20, 40, 80, 100 MHz
 - TDD (DL:UL) ratios: 7:2, 3:1, 5:4, 2:2, 2:7
- Ethernet, WiFi, and 5G connectivity
- User Equipment (UE): 6 Raspberry Pis connected to 5G Telit modems
- OAIBOX™ open-source 5G testbed, USRP X310



Ratio of rounds each node was the slowest in computation, communication, and overall round time—the straggler—over 40 MHz 5G with 2:2 TDD.

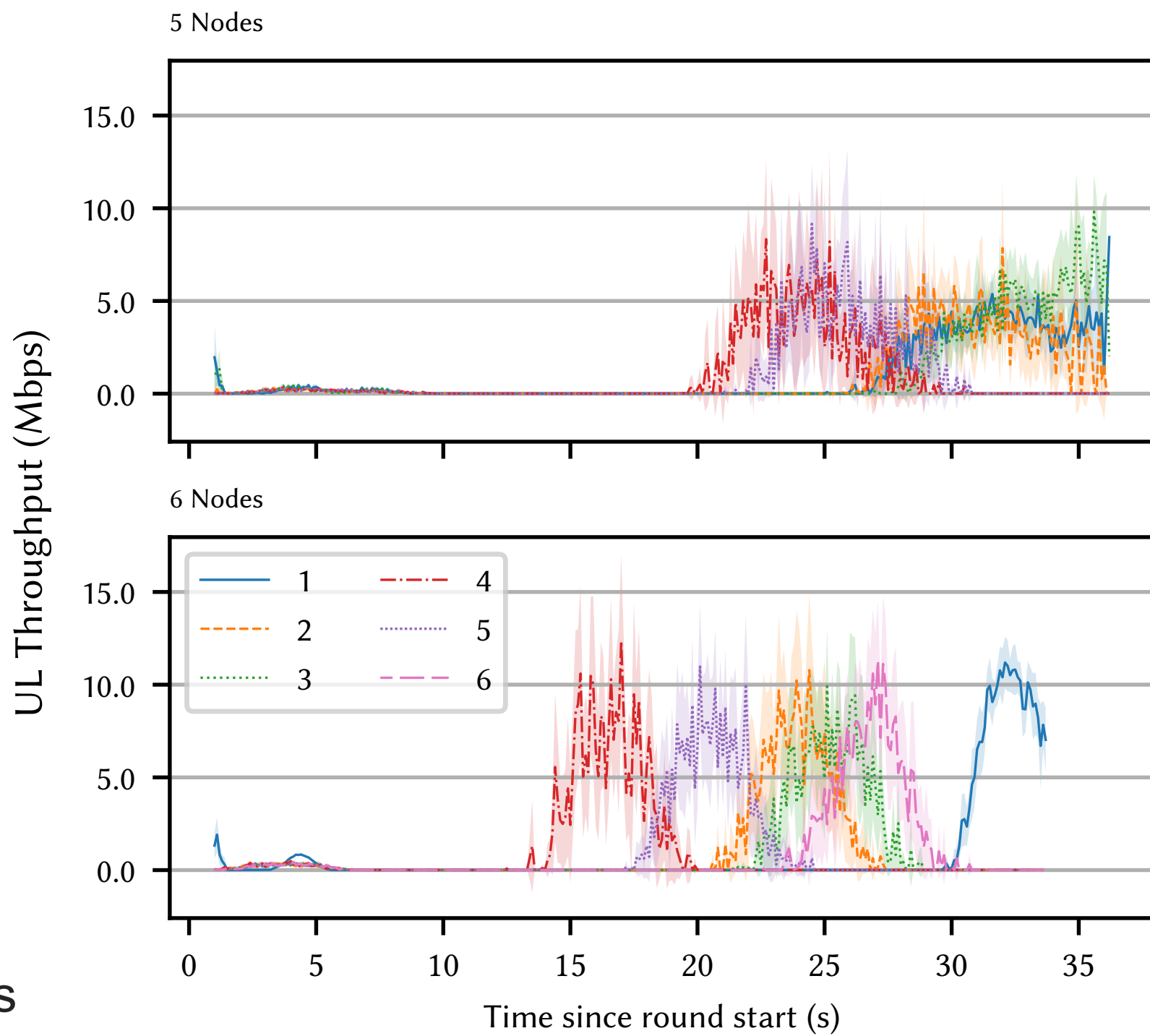
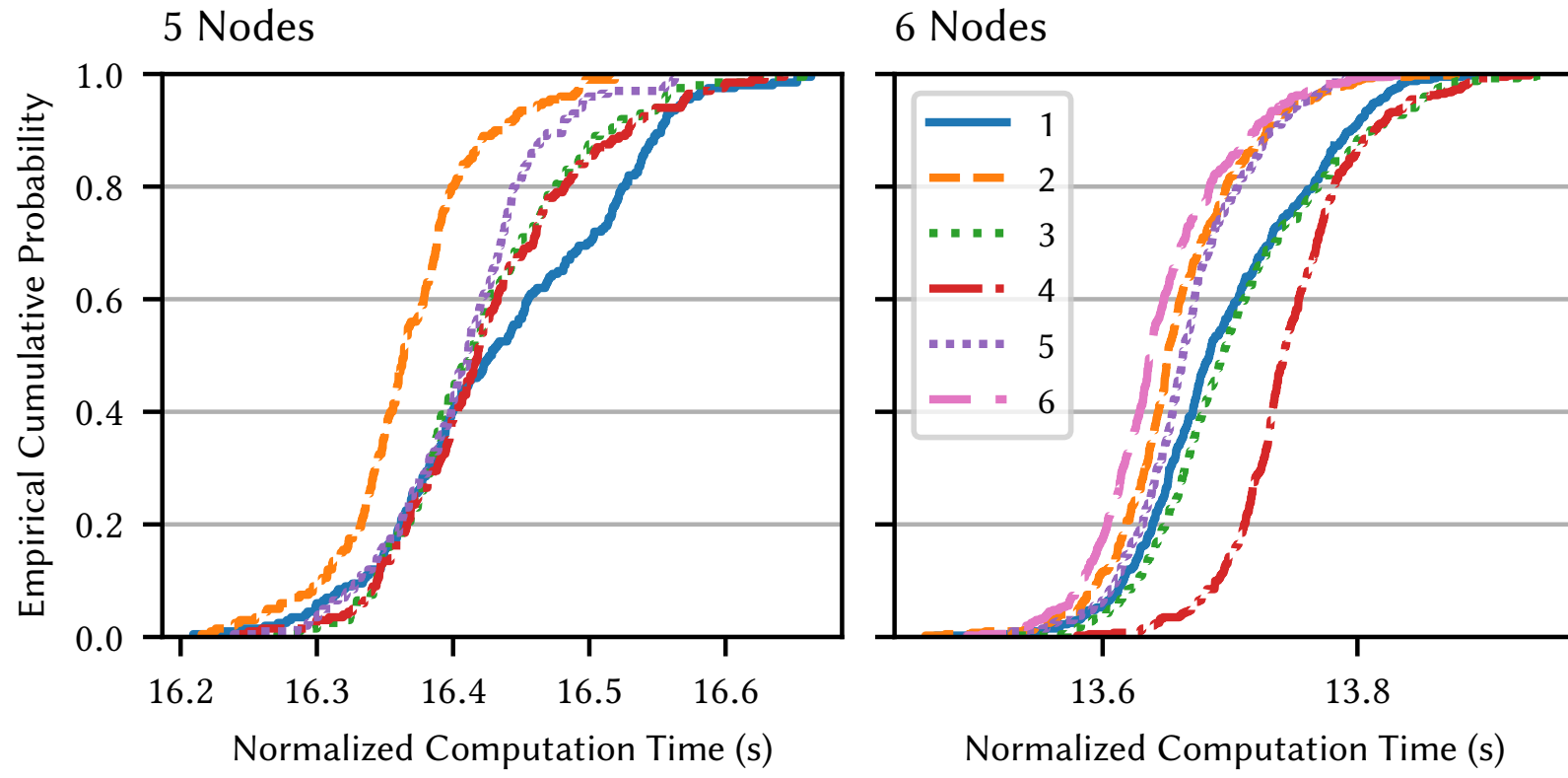
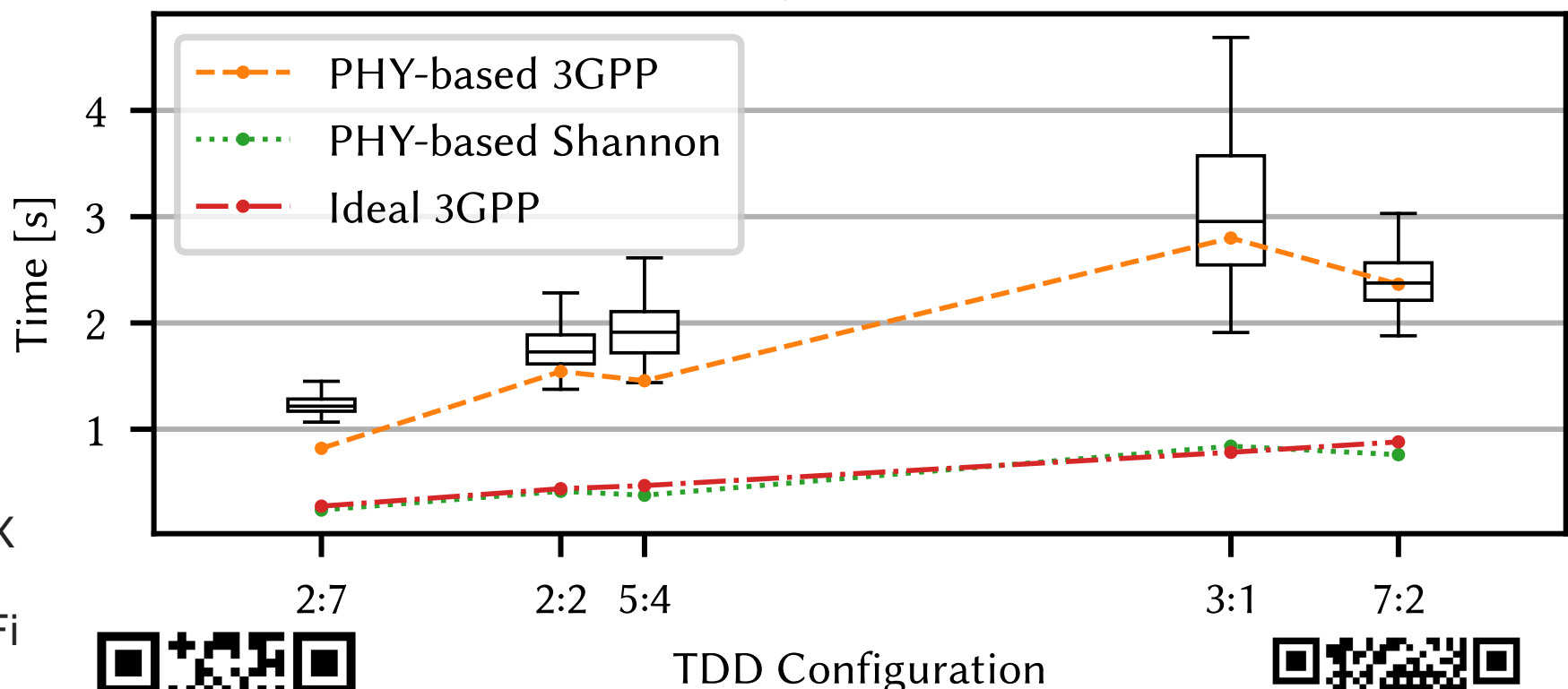
IDENTITY OF THE STRAGGLER

- Well-known in FL research that one node lags behind others, delaying everyone
- “Consistent straggler”: the same node is the straggler $\geq 95\%$ rounds
 - Always slowest in computation
 - Describes 70% of experiments, while “competing straggler” exists in 30%

THEORETICAL RATES

- Literature often assumes transmission occurs at the Shannon capacity rate [6,8]
- We compare actual UL time with Shannon capacity based on PHY-layer measurements and the 3GPP 5G standard [9]
- Measurement-based 3GPP most closely matches our results

Actual and theoretical UL transmission time across TDD configurations with six nodes over 80 MHz 5G.



Real-time UL throughput shows that the consistent straggler aligns with single-user use of channel, while the competing straggler corresponds to a shared channel.

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